

Javier Gallardo Sáenz

Mathematics PhD Student

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Education

- **PhD in Mathematics**, UBC. Supervisor: Christoph Ortner. (2026–present)
- **COSSE International MSc**, TU Delft & KTH (2024–2026)
- **Bachelor's in Mathematics, Bachelor's in Engineering Physics**, CFIS, UPC (2019–2024)
CFIS is a highly demanding, competitive dual-degree program.
- **High School Diploma**, IES Marco Fabio Quintiliano (2017–2019)
Summa Cum Laude, obtained the highest possible grade.
- **Music Studies (Guitar)**, Conservatorio Calahorra (2013–2019)

Research Experience

- **AstraZeneca, MolecularAI Dept**, Research Intern (January 2026 - August 2026)
Hierarchical, MD-assisted ML for modified siRNA activity prediction. Creation of dataset with MD properties for 6000 modified siRNA trimers.
- **IRB, Modesto Orozco Lab**, Research Intern (June–July 2024)
Flow matching for tetranucleotide dynamics; RL and statistical physics for force field parametrization.
- **UCSD, Jorge Cortes Lab**, Research Intern (Sept 2023–May 2024)
Stability and safety guarantees for Safe Gradient/Monotone Flow algorithms; robustness of dynamical systems.
- **Nostrum Biodiscovery (NBD), AI Dept**, Research Intern (June–July 2023)
Guided diffusion models for protein-ligand docking; ICLR '24 & Nature Machine Intelligence publications.
- **IREC**, Research Intern (July–Sept 2022)
P2P energy market design using game-theoretic fairness (RECOMMIT project).

Publications, Presentations, Awards & Honors

- 2026: **4YF** fellowship to the best incoming PhD students at UBC
- 2026: Master's Thesis *Molecular-dynamics-assisted hierarchical machine learning for modified siRNA activity prediction* received highest possible grade
- 2025: Paper *Informed Protein–Ligand Docking via Geodesic Guidance in Translational, Rotational, and Torsional Spaces* published in **Nature Machine Intelligence**
- 2024: Paper *GeoDirDock: Guiding Docking Along Geodesic Paths* accepted to **GEM Bio @ ICLR '24**
- 2024: Bachelor's Thesis *Extending input to state stability to closed sets: applications to robustness of anytime constrained optimization* received highest possible grade
- 2024: Presentation *Set Input to State Stability* at the MURO Lab weekly seminar at UCSD.
- 2019–2023: Multiple Honorable Mentions in Quantum Physics, Biophysics II, Solid State Physics
- 2019: Beca Ortega y Gasset (Top 100 High School Students in Spain)
- 2018: Beca Campus Científicos de Verano (Top STEM students in Spain)
- 2017: FLL Spain Best Scientific Project Award, classifying for FLL International Championship. Team captain (IES MFQ)
- 2017: La Rioja Philosophy Olympiad Winner
- 2016-2019: La Rioja Basketball League Winner. Top 25 (2016-2018) and Top 30 (2019) team in Spain. Team Captain (ABQ)

Leadership & Extracurriculars

- International Master's Ambassador, KTH (2025–2026)
- President, CMU Penafort Montserrat Laboratory (2021–2023)
- President, CMU Penafort Montserrat Debate Society (2020–2023)
- Summer Science & English Teacher (2017–2020)

Languages

- Python (Pytorch, Pytorch Geometric, JAX), C++, Julia, MD software (Gromacs, OpenMM)
- Spanish (Native), Catalan (Fluent)
- English (Cambridge C1 with level C2), lived in US, UK and Ireland
- French (4 years), student exchange in Aiguillon

Volunteering

- Asociación Baloncesto Quintiliano, social & environmental volunteering (2014–present)
- IES Marco Fabio Quintiliano, STEM outreach (2015–2019)